

Neel Viegas

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EDUCATION

- **McGill University**
Bachelor of Mechanical Engineering *Expected December 2028*
- **John Abbott College**
Science program – Dean's honor list *May 2025*
 - English – Honors certificate
- **College Charlemagne**
High School Pure sciences – International Baccalaureate (IB) Program *June 2023*
 - Certificate of exceptional performance (+90% average)

WORK EXPERIENCE

- **Pratt and Whitney Internship**
Engineering Intern for Operations Program Management *May 2026 – Present*
 - Optimization of AI database (pre-processing, integrated new data)
 - Wrote a python script for data extraction from excel and pdf files
 - Built an interactive knowledge graph app using Python, HTML and Java
 - Performing GD&T and CAD work, adding dimensions in both the 3D and 2D views in CATIA, allowing all views and dimensions to be read directly from the model without a separate drawing
 - Taught an engineer how to use SolidWorks
 - Built multiple flows (MS Power Automate) feeding an automated PM dashboard (Power BI) and Master List (MS list)
- **McGill Rocket Team**
Co-lead of the injector project *May 2026 – Present*
 - Managing injector team
 - Injector research and testing
- *Member of injector project* September 2025 – May 2026
 - 3D CADing vortex injector drilling jig in Siemens NX
 - Researching redesign of vortex injector to optimize oxidizer-fuel mixing in hybrid rocket propulsion system
- **Physics and Mathematics Tutor**
Private Tutor *September 2023 – Present*
 - Helping students learn, lab & problem-solving help
- **CI Assante Wealth Management Internship**
Assistant to financial advisor *August 2023*
 - Learned basics of: DCF models, retirement planning, stock valuation

PROJECTS

- **Thrust Analysis Tool (McGill Rocket Team)** May 2026 – Present
 - Helping run Hot Fires on a test stand rocket engine
 - Leading the creation of a python script that outputs the coefficient of combustion efficiency
- **Local AI Model (Personal Project)** June 2026 – Present
 - Used open-source AI models
 - Tested parameters for ideal process: indexing → retrieval → Augmentation & Generation
- **Automatic Curtain Opener (Personal Project)** June 2026 – Present
 - CADed gears in NX
 - Designed and assembled the circuit with an Arduino
 - Wrote a python script for timed curtain openings
- **Chair with Assisted Seating and Standing (MECH 292)** January 2026 – May 2026
 - CADed the assembly (alone)
 - Designed and assembled a circuit (team)
 - Presented the project to the class and answered design questions (team)
- **Vortex Injector Redesign (McGill Rocket Team)** August 2025 – May 2026
 - Presented several improvements to our injector based on new research
 - Updated the CAD for the full-scale and low-thrust version

SKILLS & INTERESTS

- **CAD:** NX, Solidworks, Fusion 360, CATIA
- **Coding:** Python - numpy, xlrd, openpyxl, json, os, re; HTML, Java (basics)
- **Microsoft:** Power Automate, Excel, Word
- **Teaching:** STEM teaching, CADing, problem-solving
- **Languages:** Fluent in English and French
- **Interests:** Backpacking (Casablanca to Paris summer 2025); Bouldering & Powerlifting

COMMUNITY INVOLVEMENT

- **Pratt and Whitney Intern Committee Representative** May 2026 – Present
 - Helped organize intern events with committee members
- **Refugee Language Teaching** March 2026 – Present
 - Helping refugees learn English and French on Haven
- **Club Membership** August 2025 – Present
 - Member of the McGill climbing club
- **Student Club Management**
 - Founder and leader of the Powerlifting Club (John Abbott College) *January 2024 - May 2024*
 - Co-leader of the Chess Club (John Abbott College) *September 2023 - December 2024*